
Zain Dana Harper | Systems Portfolio Brief

Seattle, Washington

<https://harperz9.github.io> | <https://github.com/HarperZ9>

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What I build

I build the hard connective tissue around model-era engineering: repository maps, evidence intake, agent orchestration, evaluation, provenance, replay, compiler and graphics systems, documentation, and public proof surfaces. The work is designed so another operator can run it, inspect it, and understand the boundary between what is shipped, experimental, measured, estimated, or still unknown.

Four capability lanes

Agent infrastructure and evaluation

- **Index 2.9.0:** repository inventory, dependency graph, documentation atlas, context packs, drift checks, and routing hints.
- **Gather 1.6.1:** provenance-aware intake across web, documents, feeds, PDFs, browser/OCR paths, and APIs.
- **Forum 1.13.0:** model-agnostic orchestration with plans, tasks, checkpoints, and replayable ledgers.
- **Crucible 1.2.0:** evaluation with a clean worker/verifier boundary.
- **Relay / Plexus / Mneme 0.1.0 source:** endpoint-aware coding-agent routing, toolchain auto-wiring, and provenance-carrying memory.

Compilers and developer tooling

- **BuildLang:** Rust compiler with typed effects, ownership/lifetime analysis, a primary C backend, shader output, editor support, and public package surfaces.
- Compiler work spans lexing, parsing, AST/IR design, type checking, code generation, effects, lifetimes, editor integration, and release tooling.

Graphics, color, and native systems

- D3D11/HLSL rendering, proxy-DLL architecture, post-processing, GPU trace validation, read-only shared-memory IPC, tone mapping, TAA, SSR, SSGI, GTAO, and volumetrics.
- A public Skyrim graphics project whose current career materials report more than 900,000 downloads.
- **Calibrate Pro 1.1:** released Windows display-calibration toolkit with a current PySide6 proposal-and-confirmation shell and evidence-labelled output. Its historical UI remains compatibility-only and is not used as current visual proof.

Documentation and operations

- API and developer documentation, implementation guides, proposals, release notes, troubleshooting, GRC and audit-support writing, and operational playbooks.
- Customer support and field-operations experience that emphasizes risk assessment, clear communication, practical handoffs, and completion.

How the work is delivered

Each serious public surface aims to include a clear purpose, install or run path, examples, tests, version and maturity language, troubleshooting, security or secret-handling posture, and a re-checkable claim trail. The same approach governs the portfolio: current native UI is shown as current; legacy UI is labelled; generated media has a reproducible renderer; demos are recorded from real workflows; and unsupported maturity claims are removed.

Best starting points

- Portfolio and current visual system: <https://harperz9.github.io/portfolio.html>
- Public repositories: <https://github.com/HarperZ9>
- Research and systems writing: <https://harperz9.github.io/research.html>
- Resume: <https://harperz9.github.io/resume.html>
- Curriculum vitae: <https://harperz9.github.io/cv.html>

Engagement fit

Strong fits include systems engineering, developer tooling, compiler and language work, AI/ML infrastructure, agent/evaluation infrastructure, real-time graphics and color, research operations, technical writing, implementation, QA, product operations, and difficult cross-domain work that needs one person to map, build, verify, explain, and ship the whole surface.

Open to full-time, contract, part-time, and project-based work. Remote preferred; onsite and hybrid work are also in scope.

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